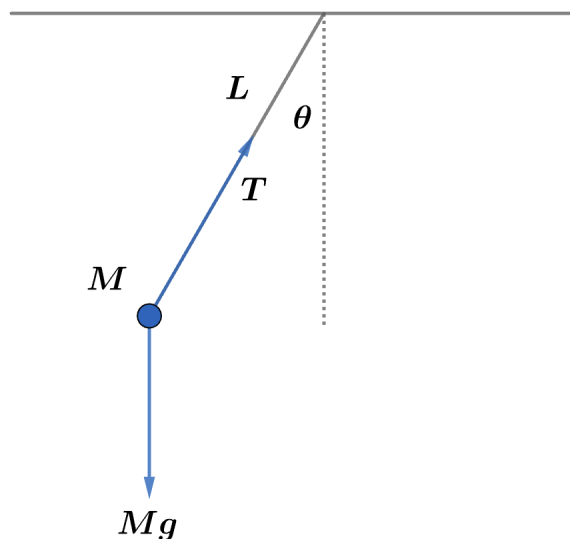


2010 F=ma Exam: Problem 9

Kevin S. Huang



Applying Newton's 2nd law, we have in the vertical direction,

$$T \cos \theta - Mg = 0$$

$$T = \frac{Mg}{\cos \theta}$$

In the horizontal direction,

$$T \sin \theta = Ma$$

Substituting in the first equation,

$$Mg \tan \theta = Ma$$

$$a = g \tan \theta$$

so the answer is $\boxed{C}$.